HOW ACCURATELY LLMS REFUSE QUESTIONS THEY DON'T KNOW? MEASURING KNOWLEDGE-AWARE REFUSAL IN LLMS

Anonymous authors
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ABSTRACT

Large Language Models (LLMs) should refuse to answer questions beyond their knowledge, which is a crucial ability for LLMs to be reliable. However, existing metrics face three challenges in quantifying this ability: **(1) inaccurate:** yielding inconsistent and biased scores; **(2) complex:** requiring heavy calibration processes; and **(3) indirect:** measuring auxiliary models rather than refusal behavior. In this work, we propose a new metric called *Refusal Index (RI)* to measure how accurately LLMs refuse questions they do not know. We define RI as Spearman's rank correlation between the refusal probability and the error probability. To calculate RI, we propose a simple and principled two-pass evaluation that estimates RI from refusal rates in two evaluation runs. Experiments across 16 models and 5 datasets show that RI quantifies a model's ability to make accurate knowledge-aware refusals. Unlike existing metrics, RI remains consistent when models exhibit different refusal rates. RI addresses an important but previously untested aspect of LLM factual evaluation. Our observations reveal a blind spot in LLM factual evaluations: while LLMs are achieving higher accuracy on factual evaluations, they are not always improving their refusal behavior. This finding highlights the need to complement traditional metrics with the Refusal Index for comprehensive factual evaluation.

1 INTRODUCTION

Large Language Models (LLMs) are increasingly used for knowledge-intensive tasks, such as long-term reasoning (Chen et al., 2025) and specialized expert domains (Wang et al., 2025; Lin et al., 2024; Mahdavi et al., 2025). Despite these capabilities, LLMs are often poorly calibrated, confidently providing incorrect answers (Huang et al., 2025). An intuitive solution is to enable models to refuse questions beyond their knowledge (Yin et al., 2023b). Recent work has explored and strengthened this ability by inducing more accurate refusals with prompting (Cheng et al., 2024; Kadavath et al., 2022b) or fine-tuning (Zhang et al., 2024; Kapoor et al., 2024). However, there lacks a reliable metric to quantify this ability.

In this paper, we focus on *how well LLMs refuse questions they don't know*, which we refer as *self-knowledge with refusals*. This assesses refusal ability in two aspects: how well a model refuses questions beyond its knowledge (avoiding *overconfidence*) and how well it avoids refusing questions it would answer correctly (avoiding *over-refusing*). Specifically, we want to measure the correlation between refusal decisions and the error probability. We find that existing metrics struggle to capture this important property:

The first limitation is that existing metrics are **(1) inaccurate** for the purpose of measuring the self-knowledge. Prior factuality evaluations (Bang et al., 2025; Wei et al., 2024; Zhang et al., 2024) report combinations of *correct answer rate* and *refusal rate* to quantify accuracy of LLM refusals. These combinations measure whether a model generates more correct answers while refusing more questions it does not know at the same time. We find that combinations of the correct answer rates and refusal rates is not sufficient to measure the self-knowledge accurately, especially when models exhibit different refusal rates. This inaccuracy is evidenced in §3.2: If the same model exhibits different refusal rates, these combinations return varying and biased scores.

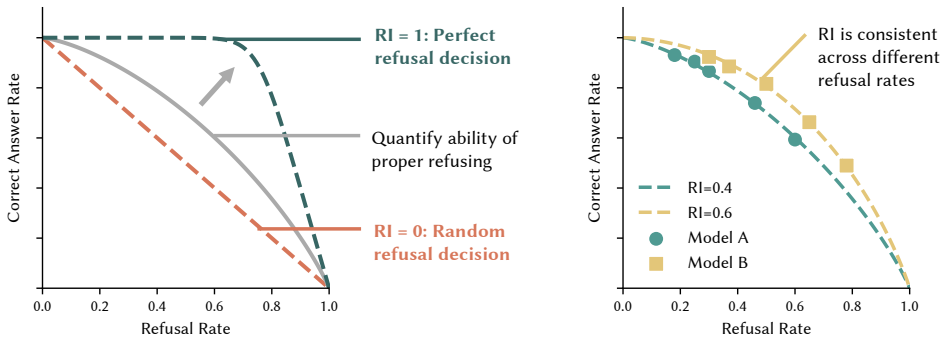


Figure 1: Illustration of Refusal Index (RI). Refusal Index quantifies a model's internal capability to refuse questions beyond its knowledge, by measuring the correlation between refusal decisions and the answer incorrectness. Left: Refusal Index models how correct answer rate drops with increasing refusal rate. Right: Empirical accuracies of the same model with different refusal rates align with the Refusal Index.

A more rigorous approach is to measure the self-knowledge with *calibration error* (Guo et al., 2017), but it is **(2) complex** and **(3) indirect**. Calibration error measures the difference between prediction probabilities and the probability of being correct, which is close to our goal. However, calibration errors are computationally **complex**. As prediction probabilities are not directly accessible from LLM answers, these metrics require a calibration process to generate prediction probabilities from model outputs (Kadavath et al., 2022b). This process can be expensive, with some methods requiring resampling up to 100 times (Wei et al., 2024) or fine-tuning language models (Ulmer et al., 2024a). Beyond complexity, calibration error is an **indirect** estimate: it measures the accuracy of the calibration process, which does not directly translate to the accuracy of an LLM's refusal decisions.

In this work, we design a metric called *Refusal Index (RI)* to address the weaknesses of previous metrics. We first formally define the Refusal Index as the Spearman's correlation between refusal probabilities and error probabilities (§2). We then introduce a principled two-pass evaluation to estimate RI directly from refusal rates. Specifically, we first evaluate a model on a question-answering dataset, collecting correct answer rates and a set of refused questions. Then, we run a second evaluation pass to regenerate answers for questions refused in the first pass. Finally, we estimate the Refusal Index with correct answer rates and refusal rates from the two evaluation passes.

We measure RI across 16 models on 5 datasets and perform extensive analyses. In §3.2, we first demonstrate how RI quantifies the capability of models to refuse questions they don't know. RI parameterizes an accuracy-refusal curve describing how correct answer rate drops as the refusal rate increases. The curve's convexity uniquely captures a model's ability to reduce false refusals. We illustrate the curve in Figure 1. Then, to empirically validate RI, we first verify that the two-pass evaluation returns consistent RI scores on the same model with different refusal rates, aligning with the accuracy-refusal curve. We then demonstrate that the RI scores from the two-pass evaluation correlate strongly with established calibration methods such as AUROC with resampling (Kadavath et al., 2022b; Ulmer et al., 2024b).

Refusal Index fills an important but overlooked gap in LLM factual evaluation. In §4, our observations reveal that (1) RI identifies models that are factually accurate but make poor refusal decisions on questions outside their knowledge, and (2) while newer models show across-the-board improvements on factual tasks, their Refusal Index does not always improve. This finding highlights the need to complement traditional metrics with the Refusal Index for comprehensive factual evaluation.

2 Refusal Index

Problem. When designing a pragmatic metric to measure LLM's ability to refuse questions beyond it knowledge, we face some challenges. First, refusals in LLMs is not fully determined by its self-knowledge, because refusal decision may be affected by model preference, risk of the questions,

or simply instructions. This precludes simply counting refusal rates for evaluating self-knowledge. For example, if we measure refusal rate on a set of unanswerable questions, model designed to have higher refusal rate can easily reach higher score. On the other hand, we want to measure refusal behavior directly without using calibrators like verbalized confidence (Xiong et al., 2023) or output from linear probes to replace refusals. Performance of calibrators may not reflect model self-knowledge with refusal. Therefore, a good metric of refusal-based self-knowledge shoud have following properties: (1) quantify model self-knowledge with refusal (2) keep consistent even when model exhibits different refusal rates. (3) derived directly from answers of blackbox LLM.

Notations. Following recent factuality evaluations such as SimpleQA (Wei et al., 2024), we focus on factual questions where model outputs can be classified as either correct or incorrect. To avoid hallucinated responses, models are allowed to abstain by producing outputs such as *"I don't have enough information..."* to avoid giving hallucinated responses. More formally, we denote the LLM as $f_{\text{LM}} : \mathcal{X} \to \mathcal{Y} \cup \{\perp\}$, where $x \in \mathcal{X}$ represents the input question, $y \in \mathcal{Y}$ represents the output answer, and $\perp$ denotes abstention (refusal). For the i-th question x_i with ground truth y_i in dataset D, we define the following indicators: $W_i = \mathbf{1}\{f_{\text{LM}}(x_i) \neq y_i\}$ for incorrect output and $R_i = \mathbf{1}\{f_{\text{LM}}(x_i) = \perp\}$ for refusal responses. We define the question difficulty as $w_i = P(f_{\text{LM}}(x_i) \neq y_i)$ as the probability of error, and $r_i = P(f_{\text{LM}}(x_i) = \perp)$ as the probability of refusal. We treat w_i and r_i as unobservable variables on $[0, 1]$.

Defining Refusal Index. Conceptually, with better self-knowledge, the model should have a higher probability of refusal when the question becomes more difficult. We propose to measure this ability with *Refusal Index*. Inspired by rank-based calibration metrics like AUROC (Niculescu-Mizil & Caruana, 2005), we define the refusal index to correlate refusal probabilities with error probabilities:

Definition 2.1 (Refusal Index)**.** The Refusal Index is the Spearman's rank-correlation between the model's refusing probability r_i and the error probability w_i.

$$\text{Refusal Index} = \rho_S = \text{Corr}(\text{Rank}(r_i), \text{Rank}(w_i)) \tag{1}$$

We use ρ_S to denote the Refusal Index for notational simplicity. To understand the intuition behind the definition, we say a model is *perfect-refusal-self-aware* when the probability of refusal is monotonically increasing with error probability, resulting the model being more likely to refuse when the question becomes more difficult.

$$P(f_{\text{LM}}(x_i) = \perp) \leq P(f_{\text{LM}}(x_j) = \perp) \iff w_i \leq d_j \tag{2}$$

Note that this differs from error-based calibration metrics like Expected Calibration Error (ECE), which quantify the absolute discrepancy between r_i and w_i. In contrast, our approach evaluates only the rank discrepancy between r_i and w_i. We define the RI as a discrimination property because it captures a more fundamental aspect of refusal-based self-knowledge. While overall refusal rates can be adjusted through input context or model preference learning, discriminative capability is more robust and consistent. In the following sections, we introduce how to estimate the Refusal Index with two evaluation passes (§2.1).

2.1 Two-Pass Evaluation

The naive way to measure RI is to acquire the probability of refusal $P(f_{\text{LM}}(x_i) = \perp)$ on questions of varying error probabilities. However, refusal probability is not directly accessible from a single evaluation run. To infer the correlation between two latent continuous variables from binary observations, we propose the a principled two-pass evaluation process. The two-pass evaluation models the refusal decision with two components: (1) treat refusal and correctness as thresholding of their respective probabilities; and (2) model their joint distribution with a Guassian copula.

Formulating the Joint Distribution. In order to estimate ρ_S from the joint distribution of the refusal and error probabilities, we use a Gaussian–copula model for the joint distribution with the correlation ρ:

$$C(u, v) = \Phi_\rho\big(\Phi^{-1}(u), \Phi^{-1}(v)\big) \tag{3}$$

where $u = F_r(r_i)$ and $v = F_e(w_i)$ are the marginal CDFs, Φ^{-1} is the standard normal quantile function, and Φ_ρ is the bivariate standard normal CDF with correlation ρ. The Φ_ρ depends only on the rank correlation, decoupled from marginal distributions of the two probabilities. In what follows we avoid modeling F_r and F_e directly and estimate ρ from R_i and W_i via maximum likelihood. We can then compute ρ_S from ρ using the relationship $\rho_S = \frac{6}{\pi}\arcsin\big(\frac{\rho}{2}\big)$.

To estimate ρ, we need to observe two binary coarsenings for each sample: R_i for the refusal probability r_i and W_i for the error probability w_i. To this end, we propose a two-evaluation pass that evaluate the model on the same dataset twice: first pass to observe refusal R_i and second pass to observe correctness W_i. In the first pass, we use a standard setup to allow model answer or refuse each question, where we classify responses into correct, incorrect, or refused. In the second pass, we update the system prompt to remove the content related to the abstention policy and only prompt the model to answer questions. We provide prompt details in §A We perform the second pass only on the subset of questions that were refused in the first pass, collecting the correctness indicators W_i' for all refused questions ($R_i = 1$).

Estimating Refusal Index. We defined the aggregated correctness indicator $\hat{W}_i = \mathbf{1}\{W_i' = 1 \text{ if } R_i = 1 \text{ else } W_i\}$ as the correctness indicator when the model answered the question. Let the empirical refusal rate be $r = \sum_{i=1}^{|D|} R_i/|D|$ and the incorrect rate be $\mu = \sum_{i=1}^{|D|} \hat{W}_i/|D|$. Under the latent uncertainty model, the pair $(R, \hat{W})$ arises by thresholding a bivariate standard normal vector (Z_R, Z_W) with correlation ρ at

$$\tau_R = \Phi^{-1}(1 - r), \qquad \tau_W = \Phi^{-1}(1 - \mu).$$

Let n_{ab} be the counts of $(R = a, \hat{W} = b)$ for $a, b \in \{0, 1\}$. The cell probabilities are

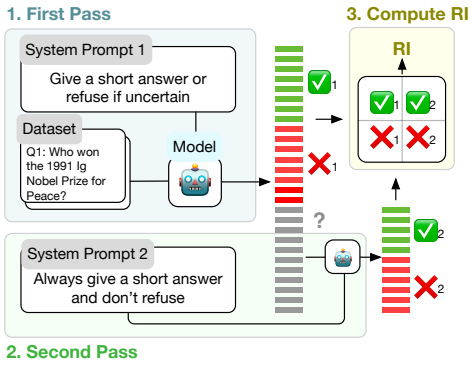


Figure 2: Two-pass evaluation process.

$$\begin{aligned} p_{11}(\rho) &= \bar{\Phi}_2\big(\tau_R, \tau_W; \rho\big) \;=\; P(Z_R > \tau_R, Z_W > \tau_W), \\ p_{10}(\rho) &= r - p_{11}(\rho), \qquad p_{01}(\rho) \;=\; \mu - p_{11}(\rho), \\ p_{00}(\rho) &= 1 - r - \mu + p_{11}(\rho), \end{aligned}$$

We estimate $\hat{\rho}$ by maximizing the multinomial log-likelihood, which are then used to compute ρ_S:

$$\hat{\rho} = \arg\max_{\rho \in (-1,1)} \ell(\rho), \quad \text{where} \quad \ell(\rho) = \sum_{a,b \in \{0,1\}} n_{ab} \log p_{ab}(\rho). \tag{4}$$

3 Experiments & Results

3.1 Experimental Setup

Models We evaluate RI on 16 open-source and proprietary models, covering different model families, sizes and release dates. Open-source models include *Gemma-3-12B* (Gemma Team, 2025), *Qwen3-32B/235B* (Qwen Team, 2025) in both think and no-think mode, *Qwen2.5-72B-Instruct* (Qwen Team et al., 2024), *Llama 3.1 70B* (Grattafiori et al., 2024), *Mistral-Large-Instruct-2411* (Mistral AI, 2024), *GLM-4.5 and GLM-4.5-Air*(GLM-4.5 Team et al., 2025) and *DeepSeek-V3-0324* (DeepSeek-AI et al., 2024). For proprietary models, we use *Claude 3.5 haiku* (Anthropic, 2024), *Claude Sonnet 4* (Anthropic, 2025), *GPT4.1* and *GPT4.1 mini* (OpenAI, 2025) and *Gemini 2.5 Flash* and *Gemini 2.5 Flash Lite* (Comanici et al., 2025). We use temperature=0.7 , top-p=0.95 for all models.

Table 1: Baseline metrics used for comparison.

Metric	Formula	Definition
Correct Answer Rate	c	Proportion of correct answers
Refusal Rate	r	Proportion of refusal answers
Correct given Attempted (C/A)	$c/(1-r)$	Correct answer rate among answered questions
F-score	$2c/(2-r)$	Harmonic mean of Correct Answer Rate and C/A
Weighted Score	$c - p(1-r)$	Weighted difference of c and r
Refusal Index	Eq. 4	Correlation between refusal decision and incorrectness

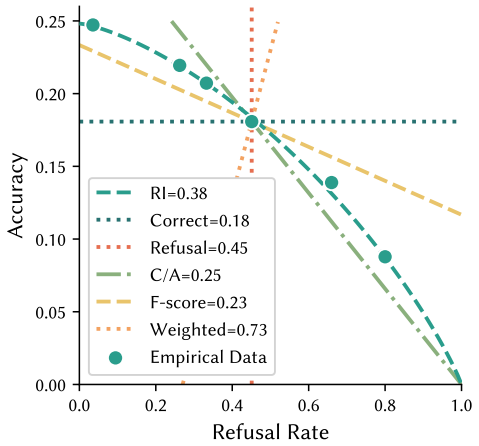
Datasets We evaluate RI on 3 scenarios: factual question answering, extrinsic hallucination detection (hallucination from training data), and intrinsic hallucination detection (hallucination from context), which all require the model's ability to make proper, self-aware refusals. For factual question answering, we use SimpleQA (Wei et al., 2024). SimpleQA contains verifiable, atomic factual questions that is even challenge for frontier LLMs. For extrinsic hallucination detection, we use PreciseWikiQA (Bang et al., 2025). PreciseWikiQA is a dynamically generated question answering dataset from snippets of wikipedia. It tests if model hallucinate its training data, given the knowledge (wikipedia) is assumed to be included in training data. We follow the Bang et al. (2025) to generate 2000 questions for evaluation. Finally, for intrinsic hallucination detection, we use the 3 datasets from FaithEval (Ming et al., 2025). FaithEval tests if the model can faithfully recall the information with a noisy context. We use all 3 datasets from FaithEval. However, as the `Unanswerable` and `Inconsistency` subset do not have ground truth, which RI needs for correct answer rate, we create a 1:1 mixed dataset of PreciseWikiQA and FaithEval to report RI.

Baseline Metrics We compare RI with 5 metrics from SimpleQA and HalluLens that are used for measuring refusal-based self-knowledge, shown in Table 1, including Correct Answer Rate, Refusal Rate, Correct given Attempted (C/A), F-score, and Weighted Score. We pick $p = 0.5$ for the Weighted Score. To produce all metrics, we follow SimpleQA to classify model output into either (In) Correct, (2) Incorrect, or (3) Not Attempted (refusal). Additionally, we further compare RI with calibration metrics AUROC with P(Answering) as an uncertainty estimation method. We follow the Wei et al. (2024) to compute P(Answering) by sampling 100 times from each question input under temperature=1 and then setting the predition probability of the question to be $1 - N_{\text{refusal}}/N$, where N_{refusal} is the number of refusal answers. The AUROC score between P(Answering) and correctness labels can be approximately served as the golden reference for Refusal Index.

Adjusting Refusal Rates We measure how the Refusal Index changes when the model exhibit different refusal rates. To this end, we use different system prompts to instruct the model to be more proactive or cautious in answering questions. Models are prompted to have different refusal tendencies without degrading its decision quality. Specifically, other than the default system prompt that make model answer all questions in the second pass, we use 4 different system prompts to evaluate each model with different refusal rates in the first pass. The details of the system prompts are shown in §A.

3.2 Refusal Rate Stability Analysis

In this section we validate Refusal Index and analyze its stability across different refusal rates. We demonstrate that: (1) the Refusal Index conceptualizes and evaluates the model's refusal-based self-knowledge with *accuracy-refusal curve*, and (2) the two-pass evaluation returns consistent RI across different refusal rates.



RI Measures Knowledge-aware Refusal with Accuracy-Refusal Curve We can use an accuracy-refusal curve to measure the accuracy

Table 2: Score stability across different refusal tendencies. We run evaluation with different refusal tendencies on SimpleQA for each model. The results are aggregated over all models in §3. Norm Δ denotes the normalized difference between most-refusal and least-refusal runs. We use $p = 1$ for the Weighted metric.

Type	**Model**	Δ_{Accuracy}	Δ_{Refusal}	$\Delta_{\text{C/A}}$	$\Delta_{\text{F-score}}$	Δ_{Weighted}	Δ_{RI}
Normalized Difference	Mistral-123B	-0.40	$+0.93$	$+0.37$	-0.16	-0.83	$\mathbf{+0.06}$
	Qwen2-35B	-0.47	$+0.95$	$+0.12$	-0.31	-0.62	$\mathbf{-0.19}$
	Qwen2.5-72B	-0.84	$+0.43$	$+0.50$	-0.60	-1.32	$\mathbf{-0.07}$
	Qwen3-32B	-0.96	$+0.54$	$+0.48$	-0.71	-1.42	$\mathbf{+0.14}$
	Average	-0.73	$+0.72$	$+0.50$	-0.50	-1.17	$\mathbf{+0.08}$
	Model	CV_{Accuracy}	CV_{Refusal}	$CV_{\text{C/A}}$	$CV_{\text{F-score}}$	CV_{Weighted}	CV_{RI}
Coefficient of Variation	Mistral-123B	0.16	0.35	0.14	0.06	0.31	**0.04**
	Qwen2-35B	0.22	0.47	0.06	0.14	0.32	**0.09**
	Qwen2.5-72B	0.35	0.17	0.19	0.26	0.53	**0.03**
	Qwen3-32B	0.35	0.19	0.17	0.28	0.51	**0.07**
	Average	0.29	0.29	0.19	0.21	0.46	**0.08**

of knowledge-aware refusal, which shows the correct answer rate at each refusal rate for a model on the same dataset. While refusing more uncertain questions reduces incorrect answers, it also leads to less correct answers due to false refusals. By varing overall refusal tendency, each model has a trade-off between correct answers and refusal rate. As shown in Figure 3, fixing any metric constant gives a unique iso-score curve in the accuracy–refusal plane, which describe the accuracy-refusal trade relationship assumed by the metric.

Compared with heuristic metrics, iso-RI curves have two desirable interpretable properties: (1) they represent sensible accuracy–refusal trade-offs matching the expected behavior of the model (see Figure 1, left). Different iso-RI curves yield the same maximum correct answer numbers when the refusal rate is 0 and reaching zero correct answers when the refusal rate is 1; and (2) RI is agnostic to maximum correct answer numbers and refusal rate, and it controls only the convexity of the curve, capturing how well a model preserves correct answers by reducing false refusals. That is, for two models with the same max correct answer numbers, the one with higher RI retains more correct answers when refusing the same fraction of questions. In contrast, heuristic metrics fail to capture this property, imposing approximately linear accuracy–refusal relationships at fixed scores. In summary, the Refusal Index does not simply reward higher accuracy or lower refusal rate, but captures the level of rank-calibration in refusal decisions, distinct from all previous metrics.

RI remains consistent across different refusal rates. To empirically validate RI, we test whether the Refusal Index remains consistent across different refusal rates. For this, we design a set of system prompts that progressively encourage caution in answering to induce different refusal rates, and then apply them to the same model on the SimpleQA dataset. Our results demonstrate that RI exhibits remarkable stability across these different refusal rates, while heuristic metrics show substantial variation. As shown in Table 2, RI is around 70% lower in variability compared to heuristic metrics. This empirical finding suggests that prompt-induced changes in refusal rate shift the refusal probability distribution without changing the correlation between refusal probability and error probability. In the appendix, we provide additional validation through goodness-of-fit tests for the Gaussian copula assumption.

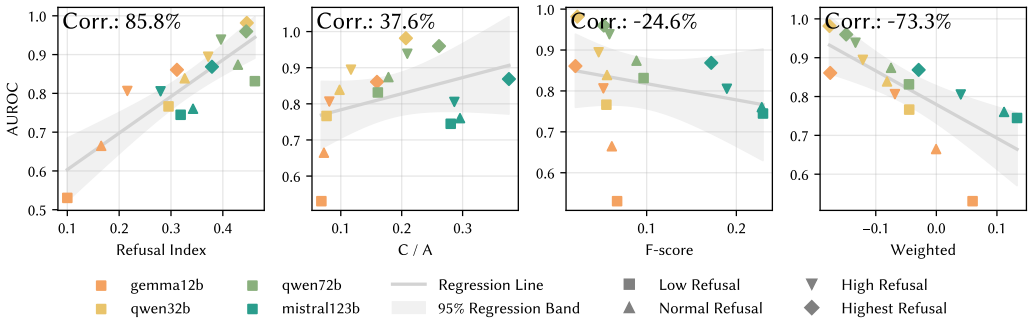


Figure 4: Correlation between factuality metrics and AUROC with P(Answering) on SimpleQA. We use $p = 1$ for the Weighted metric. RI shows the highest positive correlation with AUROC while being orders of magnitude cheaper to compute.

3.3 Alignment with Calibration Methods

RI is highly consistent with sampling-based calibration methods. One potential concern with RI is that although it is defined as the rank correlation between refusal probability and error probability, the two-pass evaluation may not faithfully reflect that correlation. To address this, we further compare RI values with AUROC with P(Answering) as an uncertainty estimation method. We choose AUROC with P(Answering) for fair comparison because it shares the same definition of uncertainty as RI. Specifically, (1) AUROC is a rank-calibration metric that only measure the discriminative ability of refusal and (2) P(Answering) is a direct and naive estimation of prediction probability of model refusals. While there exist other calibration metrics like ECE, Brier Score and various uncertainty estimation methods, AUROC with P(Answering) is the most direct measurement of the correlation between refusal and error. Figure 4 shows the results. Among the evaluated metrics, RI demonstrates the strongest positive correlation with AUROC at 88%, outperforming all other metrics. This high agreement between RI and AUROC with P(Answering) suggests that RI accurately reflects the correlation between refusal probability and error probability. Meanwhile, RI requires much smaller compute overhead compared to estimate P(Answering) with multiple sampling.

3.4 Model Ranking Stability

Finally, we study if the knowledge-aware refusal measured by RI is consistent across different models and datasets. To this end, we measure model ranking stability: does RI rank models consistently in different datasets or evaluation settings? Higher ranking stability suggests that a metric captures genuine, discriminative property of models. Specifically, we calculate Kendall's W (overall ranking agreement) and Winner Entropy (top1 consistency) on all of our 8 evaluation settings (4 refusal-varying evaluation on SimpleQA + 4 hallucination benchmarks). Additionally, to capture the ranking stability of the correlation structure between correct answers and refusals, we want to filter out the effect from solely the correct answer rate or refusal rate. We make the removal by performing a isotonic regression on the correct answer rate or refusal rate across different setups for each, then remove regressed values from each metric. We calculate the Kendall's W and Winner Entropy on the residual, effectively measuring parts of the metric that cannot be explained by correct answer rate or refusal rate alone.

RI provide stable model ranking independent of accuracy and refusal rate. As shown in Table 3, when removing the monotonic effect of correct answer rate or refusal rate, RI still shows high ranking stability, while heuristic metrics degrade to near random values. We can see that heuristic metrics like F-score and Weighted achieve very good ranking stability, but their Kendall's W and Winner Entropy drops quickly when we remove the monotonic effect from correct answer rate or refusal rate. This result suggests that while heuristic metrics like F-score has high ranking stability, they are mostly explained by correct answer rate or refusal rate. In contrast, RI retains most of its ranking stability even after removing the effect from correct answer rate and refusal rate, indicating

Table 3: Ranking stability across settings (Kendall's W and Winner Entropy).

Metric	Kendall's W ↑				Winner Entropy ↓			
	Default	−Correct	−Refusal	−Both	Default	−Correct	−Refusal	−Both
Random Value	0.25	0.25	0.25	0.25	0.61	0.61	0.61	0.61
Correct Answer Rate	0.87	0.00	**0.48**	0.39	**0.00**	1.00	0.48	1.00
Refusal Rate	0.86	0.44	0.00	0.30	0.18	0.61	1.00	1.00
C / A	0.69	**0.63**	0.40	0.37	0.33	0.61	0.48	0.61
F-score	**0.90**	0.10	0.48	0.39	0.00	**0.18**	0.48	0.52
Weighted	0.87	0.60	0.25	0.32	0.33	0.33	**0.37**	0.61
RI	0.47	0.50	0.35	**0.49**	0.47	0.33	0.61	**0.37**

that it captures the intrinsic knowledge-aware refusal of models, persisting across different evaluation settings.

4 Discussion

Can prompting models to be more cautious solve overconfidence?

Is model's self-knowledge affected by context? We can see that the RI value for PreciseWiki is similar to that of SimpleQA, while the RI value for intrinsic hallucination varies more, with larger models, especially Mistral-Large, achieving higher RI values.

Table 4: Refusal Index results on hallucination benchmarks. All results are averaged over 4 different refusal rates.

Model	Truth Available		Truth Unavailable	
	Precise-Wiki	Counter-factual	Incon-sistency	Unans-werable
Gemma-3-12b	0.36	0.56	0.22	0.12
Qwen3-32B	0.48	0.60	0.27	0.24
Qwen2.5-72B	**0.54**	0.56	0.22	0.40
Llama-3.1-70B	0.52	**0.70**	0.17	0.31
Mistral-Large	0.50	0.38	**0.34**	**0.52**
Average	0.48	0.56	0.24	0.32

What leads to better knowledge-aware refusal? Our interpretable and practical method provides several new insights into model calibration that were previously obscured by heuristic metrics. We find that the Refusal Index of even frontier LLMs is far from perfect, indicating a significant gap between their factual knowledge and their self-awareness. While external calibrators can be trained to achieve near-perfect discrimination between answerable and unanswerable questions, LLMs inherently struggle to spontaneously distinguish questions beyond their capabilities. This suggests that the internal mechanisms for generating an answer are not tightly coupled with a reliable assessment of that answer's likely correctness, leading models to confidently hallucinate on difficult questions while sometimes refusing ones they could have answered. Results are shown in Table 7. We observe that accuracy varies widely across different models, with some models achieving much higher accuracy than others. In contrast, RI exhibits only moderate variability between models, remaining relatively stable despite large differences in accuracy. This suggests that RI is less sensitive to raw accuracy and better reflects the underlying calibration and uncertainty estimation properties of the models. Furthermore, our analysis reveals that improvements in factual accuracy do not necessarily lead to a higher Refusal Index. For instance, across the 17 models evaluated on the SimpleQA dataset, we found minimal correlation between a model's overall accuracy and its Refusal Index. This decoupling implies that training for broader knowledge does not inherently confer better metacognitive skills; a model can become more knowledgeable without becoming wiser about the limits of that knowledge. Perhaps most surprisingly, we observe a trend where newer models generally perform worse on the Refusal Index, which may indicate that current training and alignment paradigms, while successful at boosting performance on many benchmarks, could be inadvertently optimizing for confident-sounding responses at the expense of genuine calibration. This trend underscores the critical necessity of including principled calibration metrics like ours directly in model development and iteration cycles to ensure that progress in capabilities is not accompanied by a regression in reliability. Overall, by moving beyond simple accuracy-refusal heuristics, our work proposes a principled, calibration-based

metric that makes factuality evaluation more reliable, interpretable, and straightforward, offering a crucial diagnostic tool for building more trustworthy AI systems without relying on computationally expensive calibration methods.

Limitations of RI. While the Refusal Index is a handy metric to measure the knowledge-aware refusal, there are limitations that practioners should be aware of. First, as the two-pass evaluation requires instructing the model to make refusals or forced answering, it requires a relatively capable model. Second, our formulation and validation of RI are focused on *knowledge-aware refusal*, which may not hold for other kinds of refusals, such as refusals for safety reasons. Finally, from both the definition and experimental results, knowledge-aware refusal is a relatively weak signal compare to other metrics like correct answer rate. Thus we recommend to use larger datasets in evaluation to get a more stable RI score. Nevertheless, RI provide a pragmatic metric for the knowledge-aware refusal, which is important but ignored by previous metrics.

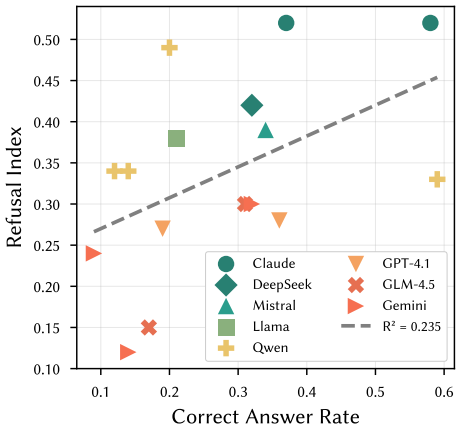


Figure 5: Scatter plot of Refusal Index vs. Accuracy on SimpleQA.

5 Related Works

Factuality of LLMs Factuality assesses an LLM's ability to generate factually correct answers. Previous evaluation methods compare LLM responses against external sources to measure factual correctness (Wei et al., 2024; Min et al., 2023; Kwiatkowski et al., 2019). Many factuality evaluations focus on measuring hallucination, the phenomenon where models generate answers that contradict available grounding information, whether provided through context or world knowledge (Bang et al., 2025). In contrast, factuality evaluation acknowledges that ground truth may not always be available to the model (Jing et al., 2025). To reduce hallucination and improve factuality, some works focus on improving the abstention policy of models by training them to appropriately refuse questions beyond their knowledge boundaries (Cao, 2024; Xu et al., 2024; Ouyang et al., 2022). Our metric does not target hallucination rate directly; instead, we evaluate the calibration level through the model's refusal behavior, quantifying the quality of these refusals.

Calibration Evaluation on Black-box Models Calibration measures the alignment between a model's output probability and its actual probability of being correct (Guo et al., 2017). It serves as a valuable metric for factuality evaluation, as it quantifies a model's self-awareness of its own knowledge (Kadavath et al., 2022a; Yin et al., 2023a; Agrawal et al., 2023). Estimating calibration for any LLM requires inferring uncertainty from text outputs of black-box models. Previous works have proposed using semantic similarity or diversity-based measures (Kuhn et al., 2023; Farquhar et al., 2024) or auxiliary models (Ulmer et al., 2024a) to estimate uncertainty, resulting in error-based or rank-based calibration metrics computed from test set outputs (Huang et al., 2024). However, these methods require training a separate calibrator for each model, which is computationally expensive and model-dependent. Our metric measures the correlation between uncertainty and difficulty, which represents a form of rank-based calibration. Our approach is lightweight since we do not estimate uncertainty directly.

6 Conclusion

In this work, we introduced Refusal Index, a novel metric to address the unreliability of existing factuality evaluations for LLMs, which are often skewed by prompt design and refusal tendencies. RI provides a principled, calibration-based measure of whether a model "knows what it doesn't know"

by evaluating the alignment between its refusal to answer and the intrinsic difficulty of a question. RI not only enables fairer model comparisons but also provides valuable insights for developing better-calibrated and more trustworthy AI systems.

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A System Prompts for Two-Pass Evaluation

To enable accurate reproduction of our results and clarify the evaluation setup described in §**??**, we provide the complete system prompts used in our experiments. We use consistent formatting instructions across all prompts to standardize outputs, and we include in-context learning examples to ensure stable model behavior and syntactically correct answers in the required format.

Second Pass System Prompt. As described in §**??**, the second pass forces models to answer questions that were refused in the first pass. We combine explicit instructions with in-context examples to enforce the output format and minimize formatting errors. Since most models we evaluate rarely refuse when given such instructions, we simply tell the model to always provide an answer. The in-context examples below help the model consistently produce the required XML-style tags. We show the complete second-pass system prompt used in our experiments:

```
Second Pass Prompt

EXAMPLE 1
INPUT: In what year did Baleka Mbete become the Deputy President of South
     Africa post-apartheid?
OUTPUT: <answer>2008</answer>

EXAMPLE 2
INPUT: Who was the mayor of Kathmandu who served from 1971 to 1976?
```

```
OUTPUT: <answer>Rajendra Man Suwal</answer>

[omitted for brevity]

EXAMPLE 10
INPUT: The book "Rhine" by Anselm Kiefer is from what year?
OUTPUT: <answer>1981</answer>

Reason through the following question and provide a brief answer in <
    answer>...</answer> at the end. You always give an answer and you
    never return <answer>UNANSWERED</answer>.

Question: {question}
```

First Pass System Prompt. Our first-pass prompts use the same design and formatting as the second-pass prompt but make small, targeted changes to the instruction wording and the mix of in-context examples to encourage appropriate refusal behavior (see §2 and §**??**). We show one complete first-pass prompt variant below; other variants differ only in how strongly they encourage refusal and the proportion of examples that show returning `<answer>UNANSWERED</answer>` when uncertain. Due to space constraints, we do not include the remaining variants in full.

```
First Pass Prompt -- Highest Refusal Tendency

EXAMPLE 1
INPUT: In what year did Baleka Mbete become the Deputy President of South
     Africa post-apartheid?
OUTPUT: <answer>UNANSWERED</answer>

[omitted for brevity]

EXAMPLE 10
INPUT: The book "Rhine" by Anselm Kiefer is from what year?
OUTPUT: <answer>UNANSWERED</answer>

Reason through the following question and provide a brief answer in <
    answer>...</answer> at the end. You are very cautious and need good
    evidence before drawing conclusions. You prefer saying you don't know
     by returning <answer>UNANSWERED</answer> rather than risking a wrong
     answer.

Question: {question}
```

As described in §3.2, these first-pass variants differ from the second pass in only two ways: (1) how strongly the instruction encourages refusal and (2) the proportion of `<answer>UNANSWERED</answer>` responses in the in-context examples. Together, these changes control the overall refusal tendency without otherwise changing the task.

B Validating the Gaussian Copula Assumption

To model the joint dependence between refusal and incorrectness under forced answering in the main text, we use the bivariate normal copula (Gaussian copula). This choice allows us to estimate the Refusal Index by capturing the correlation structure between the latent refusal score and question difficulty while remaining agnostic to the marginal distributions. However, it is important to validate whether this distributional assumption is appropriate for real model behavior.

In this section, we compare the Gaussian copula against three alternative copula families to determine which provides the best fit for modeling refusal behavior. We consider Student-t, Gumbel, and Clayton copulas as alternatives, each capturing different forms of dependence structure. We evaluate which copula family best fits the observed refusal patterns across multiple models and prompts.

Table 5: Summary of first-pass prompt variants. Only the refusal instruction and the proportion of UNANSWERED responses differ across variants; all other elements match the second-pass prompt. Ratios vary by model and dataset (see §**??**); we report relative levels for brevity.

Type	**Instruction**	**UNANSWERED ratio**
Least Refusal	You only give an answer if you are confident; otherwise you return <answer>UNANSWERED</answer>.	0.00
Moderate Refusal	You are cautious and may return `UNANSWERED` when unsure.	0.10
High Refusal	You make reasonable guesses from partial information but avoid speculation; return `UNANSWERED` if not very confident.	0.40
Highest Refusal	You are very cautious and prefer `UNANSWERED` rather than risking a wrong answer.	0.60

Evaluation Criteria. We use two criteria to evaluate copula performance: goodness-of-fit and win-rate comparisons between the guassian copula and the alternatives. Since the margins are fixed by construction in our two-pass evaluation setup, the natural goodness-of-fit criterion is the multinomial log-likelihood implied by each copula through the resulting 2×2 cell probabilities. However, since different copulas have varying numbers of parameters (e.g., Student-*t* has 2 parameters while Gaussian has only 1), we must penalize model complexity to ensure fair comparison. We complement the raw log-likelihood with the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC):

$$\text{AIC} = 2k - 2\ell(\hat{\theta}) \tag{5}$$

$$\text{BIC} = k\log(n) - 2\ell(\hat{\theta}) \tag{6}$$

where k is the number of parameters, n is the sample size, and $\ell(\hat{\theta})$ is the maximized log-likelihood. These criteria penalize more complex dependence structures, providing a principled basis for model selection. For the second criterion, we evaluate win rates by comparing how often the Gaussian copula outperforms each alternative across different model-prompt combinations. To challenge the Gaussian choice, we compare it with three standard alternatives that capture different forms of dependence. A Student-*t* copula adds a heavy-tail parameter to the Gaussian structure; a Clayton copula emphasizes lower-tail association and is asymmetric; and a Gumbel copula emphasizes upper-tail association and is also asymmetric. All candidates are fit by maximum likelihood with margins fixed at the empirical refusal and forced-answering error rates for each model-prompt combination. Win rates are computed across these individual model-prompt units to assess the relative performance of each copula family.

Experimental Setup. We systematically evaluate and compare the maximum log-likelihood for each copula family on the SimpleQA dataset. Our evaluation covers all 17 models and 4 first-pass prompts used in the main evaluation (see §**??**). For each model-prompt combination, we obtain a 2×2 contingency table with margins (r, μ) representing the refusal rate and error rate respectively. Each copula C maps these margins to cell probabilities $(p_{00}, p_{01}, p_{10}, p_{11})$, and we estimate the copula parameters by maximizing the multinomial likelihood of the observed counts as defined in Equation 7:

$$\hat{\rho} = \arg\max_{\rho \in (-1,1)} \ell(\rho)$$
$$\text{where} \quad \ell(\rho) = \sum_{a,b \in \{0,1\}} n_{ab} \log p_{ab}(\rho) \tag{7}$$

This setup isolates the copula choice while maintaining consistency with the main evaluation framework.

The results in Table 6 indicate that the Gaussian copula provides the strongest average fit and, after accounting for complexity, the best overall trade-off between parsimony and data fit. The Student-*t* copula, despite its additional heavy-tail parameter, does not improve the average log-likelihood and is uniformly worse once complexity penalties are applied. This aligns with intuition for 2×2 data with fixed margins, where heavy tails are weakly identified and tend to degenerate toward the Gaussian

Table 6: Copula comparison on SimpleQA across model-prompt combinations (ties counted as 0.5). Left panel reports mean goodness-of-fit metrics across all combinations for each copula family. Right panel reports the fraction of combinations where the Gaussian copula outperforms each alternative.

Goodness-of-Fit Metrics

Family	Log-likelihood	AIC	BIC
Gaussian	−1832.08	3666.16	3671.76
Student-*t*	−1859.14	3722.27	3733.48
Gumbel	−2086.86	4175.72	4181.32
Clayton	−2200.13	4402.26	4407.86

Gaussian Win Rates

Versus	Log-likelihood	AIC	BIC
Student-*t*	1.000	1.000	1.000
Clayton	0.676	0.647	0.647
Gumbel	0.632	0.618	0.618

Table 7: Results on SimpleQA with 95% CI.

Model	Accuracy	Refusal	C / A	F-score	Weighted	Refusal Index
Qwen2.5-72B	0.05 [0.04, 0.05]	0.74 [0.73, 0.75]	0.20 [0.18, 0.22]	0.07 [0.07, 0.08]	-0.21 [-0.22, -0.20]	0.49 [0.45, 0.53]
Qwen3-32B	0.03 [0.03, 0.03]	0.68 [0.67, 0.69]	0.12 [0.10, 0.15]	0.05 [0.04, 0.05]	-0.29 [-0.29, -0.28]	0.34 [0.28, 0.40]
Qwen3-32B-Think	0.04 [0.03, 0.04]	0.71 [0.70, 0.72]	0.14 [0.13, 0.16]	0.06 [0.05, 0.06]	-0.25 [-0.26, -0.24]	0.34 [0.29, 0.39]
Qwen3-235B	0.38 [0.37, 0.39]	0.36 [0.35, 0.37]	0.59 [0.58, 0.61]	0.45 [0.44, 0.46]	-0.27 [-0.28, -0.26]	0.33 [0.30, 0.37]
Mistral-123B	0.19 [0.18, 0.19]	0.43 [0.42, 0.44]	0.34 [0.32, 0.35]	0.23 [0.22, 0.24]	-0.39 [-0.40, -0.38]	0.39 [0.35, 0.42]
Llama-3.1-70B	0.03 [0.02, 0.04]	0.84 [0.83, 0.86]	0.21 [0.16, 0.26]	0.06 [0.04, 0.07]	-0.12 [-0.14, -0.11]	0.38 [0.28, 0.47]
GPT-4.1	0.34 [0.32, 0.37]	0.06 [0.05, 0.07]	0.36 [0.34, 0.39]	0.35 [0.33, 0.38]	-0.60 [-0.62, -0.58]	0.28 [0.19, 0.37]
GPT-4.1-mini	0.13 [0.12, 0.15]	0.31 [0.29, 0.33]	0.19 [0.17, 0.21]	0.16 [0.14, 0.17]	-0.56 [-0.58, -0.54]	0.27 [0.19, 0.34]
Claude-Sonnet-4	0.09 [0.07, 0.10]	0.85 [0.83, 0.86]	0.58 [0.52, 0.63]	0.15 [0.13, 0.17]	-0.06 [-0.07, -0.05]	0.52 [0.45, 0.60]
Claude-3.5-Haiku	0.02 [0.02, 0.03]	0.93 [0.92, 0.94]	0.37 [0.29, 0.45]	0.05 [0.03, 0.06]	-0.04 [-0.05, -0.03]	0.52 [0.41, 0.63]
Gemini-2.5-Flash	0.19 [0.17, 0.20]	0.42 [0.39, 0.44]	0.32 [0.29, 0.35]	0.24 [0.22, 0.26]	-0.40 [-0.42, -0.37]	0.30 [0.23, 0.36]
Gemini-2.5-Flash-Lite	0.08 [0.07, 0.09]	0.41 [0.38, 0.43]	0.14 [0.12, 0.16]	0.10 [0.09, 0.12]	-0.51 [-0.53, -0.49]	0.12 [0.03, 0.20]
DeepSeek-V3-0324	0.16 [0.14, 0.17]	0.50 [0.48, 0.53]	0.32 [0.29, 0.35]	0.21 [0.19, 0.23]	-0.34 [-0.36, -0.32]	0.42 [0.36, 0.49]
GLM-4.5	0.06 [0.05, 0.08]	0.79 [0.77, 0.81]	0.31 [0.26, 0.35]	0.11 [0.09, 0.12]	-0.15 [-0.16, -0.13]	0.30 [0.22, 0.37]
GLM-4.5-Air	0.05 [0.04, 0.06]	0.71 [0.69, 0.73]	0.17 [0.14, 0.20]	0.08 [0.06, 0.09]	-0.24 [-0.26, -0.23]	0.15 [0.06, 0.23]

case. The asymmetric Clayton and Gumbel copulas trail substantially on both raw fit and information criteria, though they can win occasionally on individual units.

Conclusion. We choose the Gaussian copula for two primary reasons: (1) it provides the better average fit across model-prompt combinations as evidenced by superior log-likelihood, AIC, and BIC scores; and (2) it is the simplest and most interpretable copula family, requiring only a single correlation parameter while making minimal distributional assumptions about the dependence structure. In summary, the bivariate normal copula is both simple and sufficiently accurate for the refusal–incorrectness dependence considered here. Its combination of low assumptions and competitive fit makes it a natural default for estimating the Refusal Index.

C Results on SimpleQA

D Impact of Number of Questions

The estimation of RI relies on parameters (μ, σ, and ρ) derived from the accuracy and refusal rates of our two-pass evaluation. A natural question is how the stability of RI is affected by the number of samples in the evaluation dataset. To investigate this, we assess the stability of RI by measuring its variance across subsets of the evaluation data. We create 50 randomly sampled subsets for various sample sizes (from 50 to 2000) and compute the coefficient of variation (CV) for each size, as shown in Figure 6.

The results show that while RI is less stable than other metrics with a small number of questions, its stability becomes comparable as the sample size increases. For instance, to achieve a CV of 0.1, RI requires about 25% more samples than the C/A and F-score metrics. This suggests that a slightly larger number of samples is preferable for obtaining a stable RI estimate.

E Impact of Prompt Design

We next examine how variations in prompt design affect the RI evaluation. Our experimental setup uses four distinct first-pass prompts, each with different few-shot examples and instructions, to induce

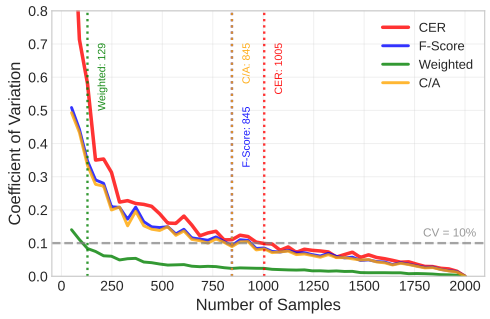

Figure 6: Coefficient of variation of RI when evaluating on subsets of the full dataset. Add significance test

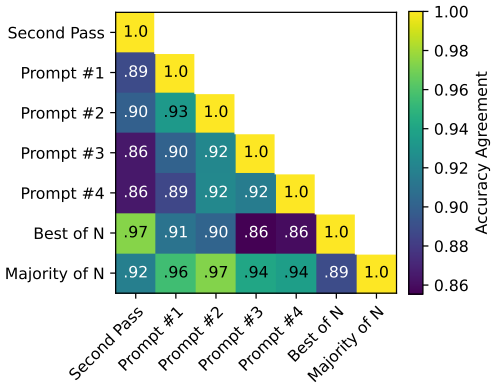

Figure 7: Accuracy agreement between different prompt strategies.

varying refusal rates. For the second pass, a single, simpler prompt is used to compel the model to answer all previously refused questions. Although these prompts are designed to produce different refusal rates, it is crucial to verify that they do not introduce confounding effects on model accuracy, which would in turn impact the RI calculation.

To assess this, we measure the accuracy agreement between pairs of prompt strategies. Agreement is calculated as the proportion of questions for which both prompts yielded the same correctness label, considering only the questions answered by both. As shown in Figure 7, the accuracy agreement between different first-pass prompts is consistently high (over 90%). This indicates that the choice of prompt strategy does not significantly alter the model's underlying accuracy on the questions it chooses to answer. Furthermore, the high agreement involving the forced-answer (second-pass) prompt validates its use for effectively estimating the model's baseline accuracy (μ).